

Ambient-energy vessel

DESIGN REPORT • SIMULATION ONLY • NO HARDWARE BUILT • 2026-08-02

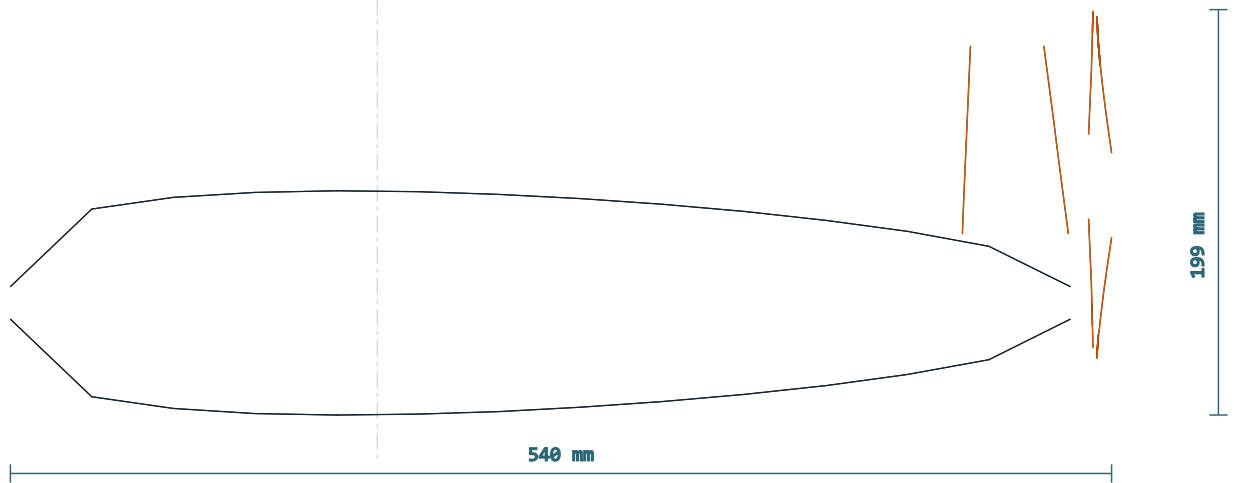
A machine that moves on wave energy alone, inside a 300 mm envelope. 432 configurations were swept through a quasi-steady hydrodynamic model and ranked by annual distance covered. This report gives the governing math, the four best designs, a bill of materials for the winner, and the land result.

DESIGNS SWEPT 432 full factorial	BEST 8,239 km / year	MEDIAN 4,639 km / year	WORST 2,487 km / year
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1 — Drawings

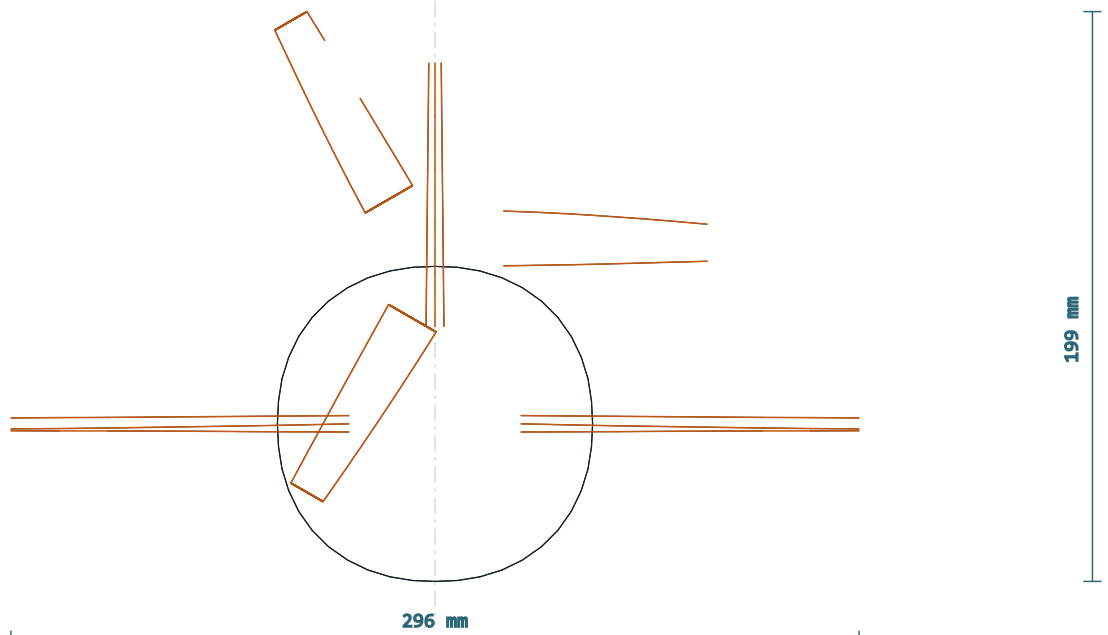
Orthographic projections taken directly from the validated mesh — silhouette and hard edges, no hand-drafting. Orange indicates moving parts. Dimensions in mm. The 2 m tether is broken out; the two assemblies are detailed at usable scale.

SUB ASSEMBLY – SIDE ELEVATION



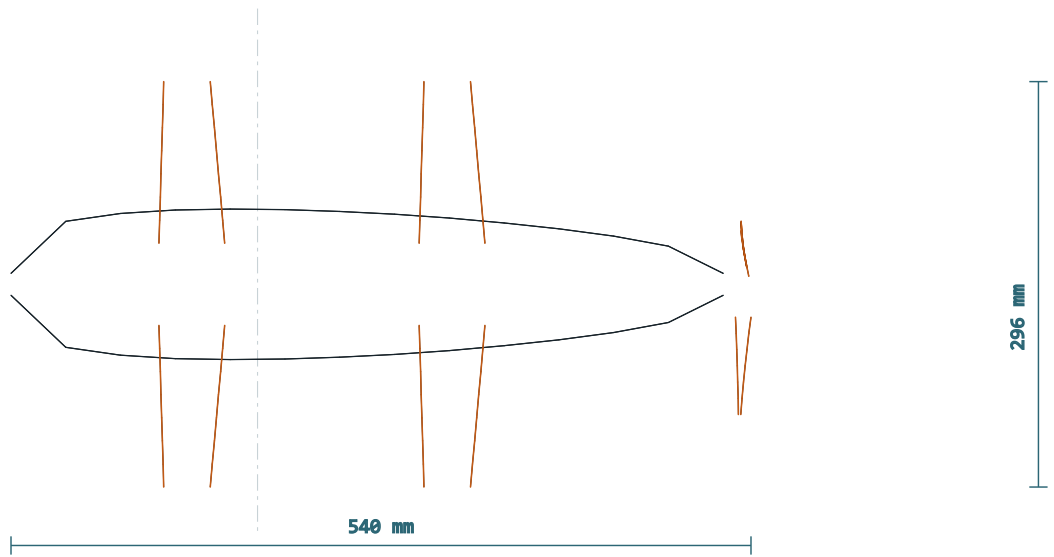
ORTHOGRAPHIC • SILHOUETTE + HARD EDGES • mm • ORANGE = MOVING PARTS

SUB ASSEMBLY – FRONT ELEVATION (looking aft)



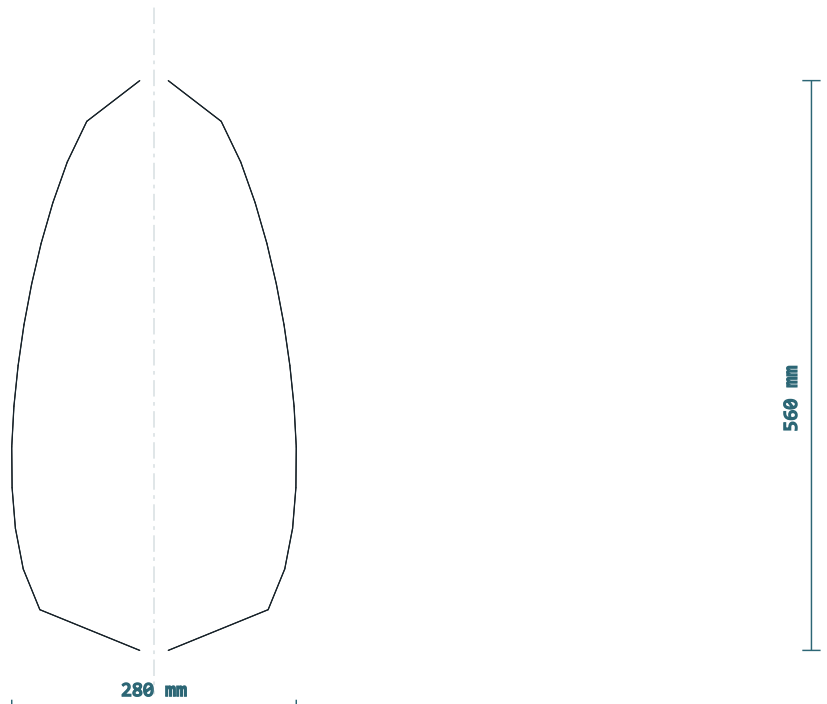
ORTHOGRAPHIC • SILHOUETTE + HARD EDGES • mm • ORANGE = MOVING PARTS

SUB ASSEMBLY – PLAN



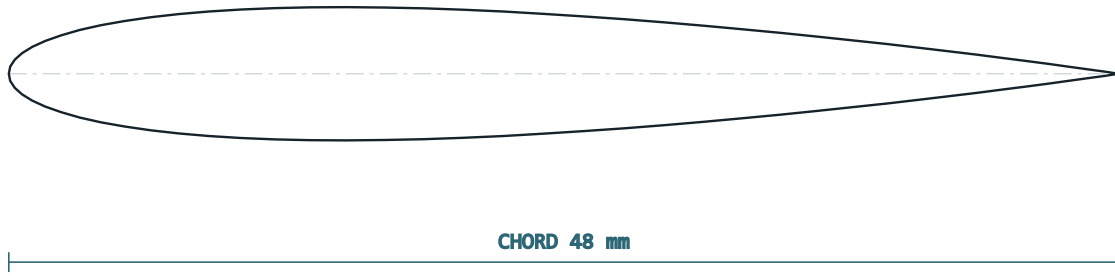
ORTHOGRAPHIC • SILHOUETTE + HARD EDGES • mm • ORANGE = MOVING PARTS

FLOAT – SIDE ELEVATION



ORTHOGRAPHIC • SILHOUETTE + HARD EDGES • mm • ORANGE = MOVING PARTS

WING SECTION – ROOT



NACA 0012 • t/c 12% • PITCH AXIS 25% CHORD • TRAILING EDGE CLOSSES EXACTLY

The section is the actual root profile lifted from the mesh — 86 points, NACA 0012, pitch axis at 25% chord. The trailing edge closes exactly, which is what makes the solid manifold.

2 — The math

WAVE INPUT

The float follows the surface, so the submerged wing is forced to heave. Heave *velocity* is the driving quantity, and it depends on wave height *and* period:

$$w(t) = A \cdot \omega \cdot \cos(\omega t) \quad A = H/2, \quad \omega = 2\pi/T$$
$$w_{\max} = \pi \cdot H / T \quad \leftarrow \text{steepness, not height}$$

THRUST FROM INCLINED LIFT

A wing moving forward at U while heaving at w sees flow arriving at angle θ . Lift acts perpendicular to that inclined flow, so it tilts forward and gains a thrust component:

$$\theta = \text{atan}(|w| / U) \quad \alpha = \max(\theta, \theta - \beta) \quad \beta = \text{mechanical stop}$$
$$V = \sqrt{U^2 + w^2} \quad q = \frac{1}{2}\rho V^2 S$$
$$T = q \cdot [C_L(\alpha) \cdot \sin\theta - C_D(\alpha) \cdot \cos\theta]$$

positive only when $\tan\theta > C_D/C_L$

Both halves of the cycle push: the wing flips to its opposite stop and θ reverses with it.

VERTICAL BALANCE — WHAT LIMITS HEAVE

The wing's force normal to the flow resists the heave. On the upstroke the float's reserve buoyancy must carry it; on the downstroke only ballast pulls the sub down. When the wings load harder than that budget, the sub heaves *slower than the wave*:

$$F_z = C_L \cdot \cos\theta + C_D \cdot \sin\theta \quad \text{per wing, resists heave}$$
$$n \cdot F_z \leq (B_{\text{float}} - W_{\text{sub}}) \quad \text{up} \quad | \quad \leq W_{\text{sub}} \quad \text{down}$$

WAKE DEFICIENCY

A flapping foil never develops full steady lift. Theodorsen's function, via the R.T. Jones rational approximation, with reduced frequency k :

$$k = \omega \cdot c / (2V) \quad C(k) = 1 - 0.165/(1 - 0.0455i/k) - 0.335/(1 - 0.3i/k)$$

RESISTANCE AND CLOSURE

$$C_F = 0.075 / (\log_{10} Re_L - 2)^2 \quad \text{ITTC-57}$$

$$D = \frac{1}{2} \rho U^2 \cdot S_{\text{wet}} \cdot (C_F \cdot k_{\text{form}} + C_R(Fr))$$

solve $\bar{T}(U) = D(U)$ **for** U $\bar{T} = \text{cycle mean}$

THE GOVERNING LIMIT

$$U_{\text{max}} = w_{\text{heave}} / \tan(\beta_{\text{stop}})$$

Once U rises enough that θ falls below the stop angle, the wing weathervanes flat, $\alpha \rightarrow 0$, lift vanishes — and profile drag remains. Thrust goes to exactly zero. Every equilibrium in this report sits just under this ceiling.

3 — Top four designs

Full factorial over wing count, span, chord, stop angle and float size. Ranked by distance covered over an occurrence-weighted temperate coastal year.

#	WINGS	SPAN	CHORD	STOP	FLOAT	KM/DAY	KM/YEAR	UNDER WAY	KT @ MOD
1	8	140 mm	60 mm	20°	15 kg	22.6	8,239	65%	0.89
2	8	140 mm	60 mm	16°	15 kg	22.2	8,118	65%	0.84
3	8	115 mm	60 mm	20°	15 kg	22.2	8,109	65%	0.87
4	8	140 mm	60 mm	25°	15 kg	22.0	8,042	65%	0.94

All four converge on the same shape: **maximum wing count, maximum chord, smallest float**. The four differ only in stop angle and span, and they differ by under 2.5%.

WHAT ACTUALLY DRIVES THE RANKING

WING COUNT	MEAN KM/YEAR	STOP ANGLE	MEAN KM/YEAR
2	4,059	12°	4,684
4	4,902	16°	4,838
6	5,203	20°	4,938
8	5,278	25°	4,982

THIS CORRECTS AN EARLIER FINDING

An earlier analysis concluded "you cannot out-wing it" — that wing area barely matters because speed is capped by U_{max} . That held wetted area *fixed* while adding wings, which is not physical. With drag charged honestly for every wing added,
GOING FROM 2 TO 8 WINGS STILL GAINS 30%
(4,059 → 5,278 km/yr).

Both statements are true and they are not in conflict. Wings do not raise the ceiling. What they do is **LOWER THE SEA STATE AT WHICH YOU REACH IT**
— and since roughly 70% of a temperate year is flat-to-light, that is where the annual distance is actually won. Stop angle, by contrast, spans only 6% across its whole range; it is nearly a free choice.

4 — Bill of materials

Design #1. Costs are deliberately omitted — no sourcing was done, and any figure here would be invented. Quantities, materials and make/buy are real.

ITEM	QTY	MATERIAL / SPEC	SOURCE
FLOAT — SURFACE			
Spar float hull	1	15 kg displacement, 280 mm dia	Printed / rotomold
Solar panel	1	Semi-flexible, marine	COTS
Electronics enclosure	1	IP68	COTS
GNSS + antenna	1	—	COTS
Satellite modem	1	Iridium SBD	COTS
Nav light	1	LED, all-round white	COTS
TETHER			
Umbilical	1	2 m, Kevlar-reinforced, 4-conductor	COTS
Potted termination	2	With strain relief	Fabricate
SUB — PROPULSION			
Sub body	1	110 mm dia × 520 mm	Printed
Wing	8	NACA 0012, 140 mm span, 60 mm chord, 4° washout	Printed
Wing hinge assembly	8	316 SS axle, PTFE bushing, 20° stops	Fabricate
Rudder	1	NACA 0012, 52 mm root chord	Printed
Rudder actuator	1	Waterproof servo — only actuated axis	COTS
Ballast	1 set	Trim to 12 N net negative	Fabricate
HOUSEKEEPING POWER			
Current spindle blade	3	190 mm dia rotor	Printed
Spindle generator	1	Low-RPM 3-phase + rectifier	COTS
Battery	1	~100 Wh — carries 30 mW across calms only	COTS
AVIONICS			
Flight computer	1	Swarm edge node	COTS
IMU + magnetometer	1	9-DOF	COTS
Depth sensor	1	Pressure, 0–30 m	COTS

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ITEM	QTY	MATERIAL / SPEC	SOURCE
CONSUMABLES			
Fasteners, seals, anodes	lot	316 SS, EPDM O-ring, Zn	COTS

13 printed parts (float, sub, 8 wings, rudder, 3 spindle blades — 14 with the rudder). Every printed part was validated closed and manifold: 0 open edges, 0 non-manifold, 0 degenerate facets, Euler $\chi = 2$.

5 — Land: no machine found

An earlier draft of this report claimed a land crawler at 148 m/day. That claim is withdrawn. It was arithmetic on an incoherent premise.

WHAT THE SOIL MODEL DOES GIVE

Soil has no usable geothermal gradient at probe scale — 25 K/km means a 3 m probe sees 0.075 K. The usable gradient is the **thermal wave lag**: surface temperature swings, soil damps and delays it with depth, and the difference between two depths drives a thermoelectric couple.

$$T(z, t) = \bar{T} + A \cdot e^{-z/d} \cdot \cos(\omega t - z/d) \quad d = \sqrt{2\alpha/\omega}$$
$$\alpha = 0.5 \times 10^{-6} \text{ m}^2/\text{s} \rightarrow d = 2.241 \text{ m annual, } 0.117 \text{ m diurnal}$$

A shallow 0→0.3 m junction pair beats a deep 0→2.5 m pair **3.5 to 1** — thermal resistance scales with separation and heat *flow* is what converts, not ΔT . Dual pair, 1 m² collector: **23 mW**. That is a real and useful result.

WHY IT CANNOT MOVE

- **The harvest requires a buried static probe.** One junction must sit ~300 mm underground and the other at the surface. A vehicle that moves cannot have a probe in the ground. It would have to stop, drill in, harvest for a day, extract, and move — that is a machine that plants itself, not a rover.
- **The 23 mW assumed a 1 m² collector.** A sub-foot vehicle cannot carry one. At rover scale the harvest is nearer 1 mW — another 20× worse.
- **Even the optimistic figure was 58× short.** Crawling a 5 kg rover at 0.10 m/s over soft ground (C_{rr} 0.15, drivetrain η 0.55) needs 1.34 W against 23 mW available.

LAND VERDICT

No land mover was found. The soil-thermal work stands as a **STATIC**

sensor node and nothing more. Moving on land with ambient energy requires wind or photovoltaic — neither of which was modelled here. Note also that the soil thermal wave is itself solar-driven; describing it as “not solar” would be a distinction without a difference.

6 — What is not true

- **Nothing was built.** No hardware, no tank test, no sea trial, no measured data of any kind. Every number in this report is simulation output.
- **No site data is real.** Tidal constituents, soil profiles and the sea-state occurrence mix are illustrative parameter sets, not surveys.
- **Drag is a correlation stack, not a solved boundary layer.** Section C_D reads roughly $1.2\text{--}1.5\times$ conservative.
- **No structural analysis.** Whether a printed wing survives hinge loads at 5.3 million cycles per year is unknown and may drive the part out of print.
- **Stop angles below 15° are unmodelled** — hinge loads, control authority and tether snatch are absent.
- **Self-intersection is not checked.** Mesh topology is verified closed and manifold; a sufficiently twisted loft could pass and still be unprintable.
- **This never ran in the swarm.** No ticket, no bot-node, no manifest. Built by hand.

WHAT IS HARD-CHECKED

- 378 unit assertions across 9 spec files; typecheck clean on both project configs.
- Betz limit holds — ideal rotor peaks at C_p 0.5776 against $16/27 = 0.5926$, approached from below, never crossed across 24 tip-speed ratios.
- Panel method reproduces thin-airfoil theory: $C_L(0) = 5.7\times 10^{-15}$ symmetric, lift slope $\rightarrow 2\pi$ per radian, NACA 4412 zero-lift -4.27° vs published $\approx -4.1^\circ$.
- Geometry and CAD export survived 17 of 17 deliberate mutations.
- 13 defects found by adversarial review and fixed — two of which had certified seasonally dead designs as perpetual.